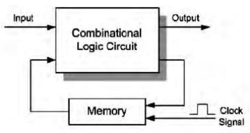
MODULE 8

COUNTERS

Recap: Sequential Circuit

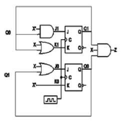


- General sequential circuit that consists of a combinational logic section and a memory section that is flip-flops.

- Memory elements are devices capable of storing binary information.
- Binary information stored at any given time defines the state of the sequential circuit.
- The input and the present state of the memory element determines the output.
- A sequential circuit is specified by a time sequence of inputs, outputs, and internal states.

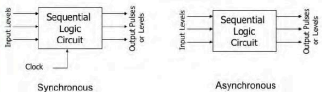
How does sequential circuits look like?

Below is a sequential circuit with two JK flip-flops. There is input X and output Z. The values of the flip-flops (Q1Q0) form the state, or the memory, of the circuit. The flip-flop outputs also go back into the primitive gates on the left. This matches the abstract sequential circuit diagram in the slide.

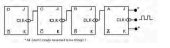


Types of Sequential Circuits

- Asynchronous and Synchronous.
- Asynchronous sequential circuits do not use a common clock signal. The circuit operation is based on message (signal) passing from one stage (e.g. circuit) to another.
- Synchronous sequential circuits use clock inputs to synchronize the circuit operation.



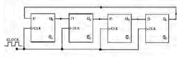
For asynchronous counters , not all flip flops are hooked to the same clock.



Asynchronous Counter

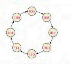
Asynchronous counters, also known as ripple counters, are not clocked by a common pulse and hence every flip-flop in the counter changes at different times. The flip-flops are usually clocked by the output pulse of the preceding flip-flop.

Synchronous counters have a common clock which results in all the flip-flops being triggered simultaneously.



Synchronous Counter

Counters



- Counters are sequential circuit that goes through a prescribed (predefine) sequence of states upon the application of input pulses.
- Sequence of states in a counter may follow a binary count or any other sequence.

- Counters can retain an output state after the input condition which brought about that state has been removed.
- Counter can have more than two states
  - value of a state is expressed as a multi digit binary number
  - the number `1's and `0's are usually derived from the outputs of internal flip-flops that make up the counter.
- The number of states a counter may have is limited only by the amount of electronic hardware that is available.
- The changes from one state to another is controlled by the input pulses (count pulses). It may be clock pulses or originated from an external source and may occur at prescribed intervals of time or at random.

- Counters are generally made up of flip-flops and logic gates.
  - Main types of flip-flops used in designing counter are J-K, T or D flip-flops.

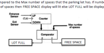
Uses of Counters

- To count the number of times that a certain event takes place; the occurrence of event to be counted is represented by the input signal to the counter (see figure below).
- To control a fixed sequence of actions in a digital system
  - e.g gate A must be open before gate B.
- To generate timing signals
  - e.g a bomb can only be detonated 20 second after a button is pressed
- To generate clocks of different frequencies.
  - By dividing the frequency, we can get a different clock frequencies from the same source

An application of a counter:

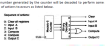
1. **Parking Space**

   There are sensors at the entrance and exit of the parking lot. A car at the entrance will increment the counter and car exit from the parking lot will decrement the counter. The current number of car in the parking lot will be compared to the Max number of spaces that the parking lot has. If number of car less than max number of spaces then FREE SPACE display will lit else LOT FULL will be displayed.



2. **Action sequencer**

   A counter can generate a number 0 to 5. Every positive edge of a clock the number will change to the next number. Every number generated by the counter will be decoded to perform some action. This will allow a sequence of actions to occurs as listed below.



Count Sequence

- A counter may count up or count down or count up and down depending on the input control.
- Because of limited word length, the count sequence is limited.
  - For an n-bit counter, the range of the count is $[0, 2^n-1]$. The count sequence usually repeats itself.
- When counting up, the count sequence goes in this manner: $0, 1, 2, \ldots 2^n-2, 2^n-1, 0, 1, \ldots$etc.
- When counting down the count sequence goes in the same manner: $2^n-1, 2^n-2, \ldots 2, 1, 0, 2^n-1, 2^n-2, \ldots$ etc.

A counter can be constructed by an asynchronous circuit or by a synchronous circuit. With an asynchronous circuit, all the bits in the count do not all change at the same time. With a synchronous circuit, all the bits in the count change synchronously with the assertion of the clock.

A counter may count up or count down or count up and down depending on the input control.

3-bit Up Counter	3-bit Down Counter
000	000
001	111
010	110
011	101
100	100
101	011
110	010
111	001

The complement of the count sequence counts in reverse direction. If the uncomplemented output counts up, the complemented output counts down. If the uncomplemented output counts down, the complemented output counts up.

3-bit Up Counter	Complement of the Count
000	111
001	110
010	101
011	100
100	011
101	010
110	001
111	000

Characteristic of Counter

- Counter can be described by the following characteristics:
  - Asynchronous or synchronous
  - Counting sequence
  - Modulus (MOD)
  - Whether the counter repeat the counting sequence

- Asynchronous or Synchronous
  - ✓ Whether the clock come from the same source or not
    - ☐ Synchronous – all clock from the same source
    - ☐ Asynchronous – clock for each flip-flop comes from a different source

- Counting sequence: Up or Down or Up-Down
  - ✓ Up or Down – unidirectional – the count sequence only one way.
  - ✓ Up-Down – bidirectional – the direction of the count can be change during operation.

- Modulus (MOD): the number of state that the counter can have.
  - ✓ e.g. MOD 4 (4 states with the following state 0, 1, 2, 3), MOD 9 (0, 1, 2, … , 8).
  - ✓ the maximum count usually 2^N- 1, where N is the number of the FF, or for the truncated sequence the max count will be < 2^N- 1.

    E.g. MOD 9 (0, 1, 2, …, 8) is truncated sequence.

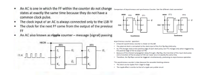
- Whether the counter repeat the counting sequence:
  - ✓ Recycle – if the counter goes back to the first count after the maximum count
  - ✓ Saturated – the counter repeat the maximum count if count up or repeat the minimum if count down.

Module Content

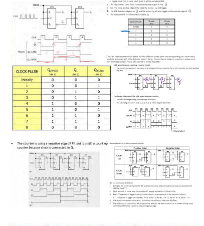
categories Asynchronous and Synchronous counter. In both categories the counters can be differentiated by the following criterion:

1. Clock Trigger: Positive edged or Negative edged
2. Counts: Binary, Decade
3. Count Direction: Up, Down, or Up/Down

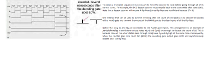
- ASYNCHRONOUS COUNTER
  - 2 bit / 3 bit / 4 bit / BCD Decade



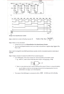
Count Up



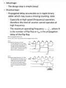
Asynchronous Decade Counter



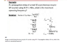
Asynchronously Clocked Modulus-12



Advantage and Disadvantage of AC (1)



Advantage and Disadvantage of AC (2)

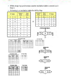


- DESIGN THE SYNCHRONOUS COUNTER

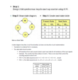
- SYNCHRONOUS COUNTER
  - 2 bit / 3 bit /4 bit / BCD Decade

Synchronous Counter Design

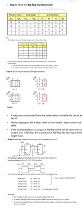
F/F Excitation Table



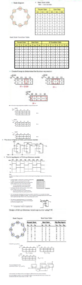
Design 2-Bit Synchronous Counter



… 2-Bit Synchronous Counter

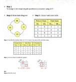


Design 3-bit Synchronous Counter

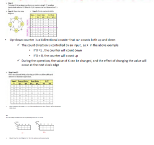


- UP/DOWN COUNTER

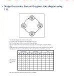
2-Bit Down Synchronous Counter



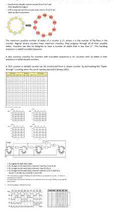
Up-Down Synchronous Counters



Counter for Arbitrary Sequences



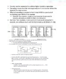
Synchronous BCD Decade Counter



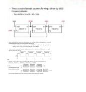
Recap… Modulus

- CASCADED COUNTER

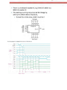
Cascaded Counter



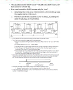
Modulus-1000 Counter



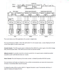
74HC163: 4-Bit Synchronous Binary Counter



Cascaded Counter with Truncated Sequence

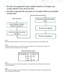


Counter Applications: Digital Clock

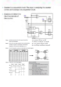


- COUNTER DECODING

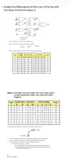
Analysis of Sequential Circuits



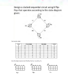
Method 1

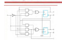


Method 2



Example: Design



… Example: Logic Diagram for Sequential Circuit with D Flip-Flops